



M002 Week 7 — English Worksheet

Topic: Multi-Lever Puzzles · **Course:** Maze Madness · **Time:** about 45 minutes

This week the maze has **multiple levers** that must be flipped in the right order. You will also wrap your "move and flip" pattern into a **function** to keep the code clean.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
function flip_next_lever:
    while no lever ahead:
        move forward
        interact ahead

flip_next_lever
flip_next_lever
flip_next_lever
```

How many levers does the agent flip? Why use a function?


```
function flip_next_lever:
    while no lever ahead:
        move forward
        interact ahead

flip_next_lever
turn right
flip_next_lever
```

Between the two function calls, the agent turns right. Why might that matter?


```
flip lever B  
flip lever A
```

The maze needs lever A flipped FIRST, then lever B. What happens with this order?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to flip **3 levers** in a row. The code repeats the same pattern 3 times — turn that into a function.

```
while no lever ahead:
    move forward
    interact ahead

while no lever ahead:
    move forward
    interact ahead

while no lever ahead:
    move forward
    interact ahead
```

Write the fixed code using a function:

Why is the function version better?

Bug B — Lever **A must be flipped before lever B**, otherwise the door stays closed.

```
flip_next_lever    # lever B
turn right
flip_next_lever    # lever A
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The function should **only flip a lever** — not turn or move after. Right now the function does too much.

```
function flip_next_lever:
  while no lever ahead:
    move forward
  interact ahead
  turn right
  move forward
```

Hint: keep the function small. Move the turn and the extra move outside.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. The maze has **multiple levers** that must be flipped in the correct order. Use **at least one function** in your solution. When you finish, come back here.

Send a photo or video of the agent reaching the end, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I planned the order of levers as ...

My function ... does ...

I called the function ... times.

The order mattered because ...

The hardest part was ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **lever**, **function**, **call**, **order**, **interact**.

1. Show all the levers in the maze and say the order to flip them.
2. Run your code and follow the agent through.
3. Read your function and the calls out loud and say what each one does.
4. Show one bug you hit and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.